

PQC Adoption Gap in 2026: Browser Readiness vs. Server-Side Lag

Day 15 | Category: PQC Adoption & Industry Status

Date: 2026-07-24 | **DAILY MONITORING**

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The Two Sides of the TLS Handshake

As of mid-2026, the transition to Post-Quantum Cryptography (PQC) for securing internet traffic via TLS presents a significant imbalance. While major web browsers have proactively integrated hybrid PQC key exchange mechanisms, the vast majority of web servers are yet to follow suit. This creates an adoption gap where the client-side is ready for a post-quantum future, but the server-side is lagging, leaving most web connections protected only by classical cryptography.

This gap is critical because of the "Harvest Now, Decrypt Later" (HNDL) threat. An adversary can record encrypted traffic today and decrypt it years later once a sufficiently powerful quantum computer is available. Enabling hybrid PQC now is the only defense against this long-term threat.

Current State of PQC Adoption (July 2026)

The data reveals a clear disparity between the readiness of browsers and the configuration of web servers.

Browser Support: Strong and Proactive

All major web browsers have enabled hybrid PQC key exchange by default, combining a classical algorithm (like X25519) with a post-quantum one (ML-KEM, specifically Kyber768).

Browser	PQC Key Exchange	Default Enabled	Hybrid Algorithm
Chrome 124+	✓	✓	X25519Kyber768
Firefox 128+	✓	✓	X25519Kyber768
Safari 18+	✓	✓	X25519Kyber768

Edge 124+	✓	✓	X25519Kyber768
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Server Support: Lagging Significantly

Server-side adoption is where the bottleneck lies. While the largest tech companies have deployed PQC, the broader internet is slow to catch up.

- **Top 100 Websites:** Approximately 40% support hybrid PQC.
- **Top 1 Million Websites:** Only around 8.6% support hybrid PQC.
- **Internal Enterprise Services:** Estimated at less than 2%.

This lag is often due to reliance on older software, lack of awareness, or concerns about the performance overhead and stability of new cryptographic primitives. However, with modern libraries like OpenSSL 3.2+, enabling it is often a simple configuration change.

Enabling PQC on a Web Server

For system administrators, enabling hybrid PQC on a modern server is straightforward. For example, using OpenSSL 3.2+:

NGINX:

```
ssl_ecdh_curve X25519Kyber768:X25519:P-256;
```

Apache:

```
SSL0OpenSSLConfCmd Curves X25519Kyber768:X25519:P-256
```

Daily Quiz

1. What is the "Harvest Now, Decrypt Later" (HNDL) threat?
2. What specific hybrid algorithm has been standardized by major browsers for PQC key exchange in TLS?
3. Besides browser and server support, what other key component of the web ecosystem needs to adopt PQC for end-to-end quantum resistance (hint: it involves identity verification)?